Analytics & Big Data

Session 3: Analytical data architecture

Prof. Dr. Gerit Wagner

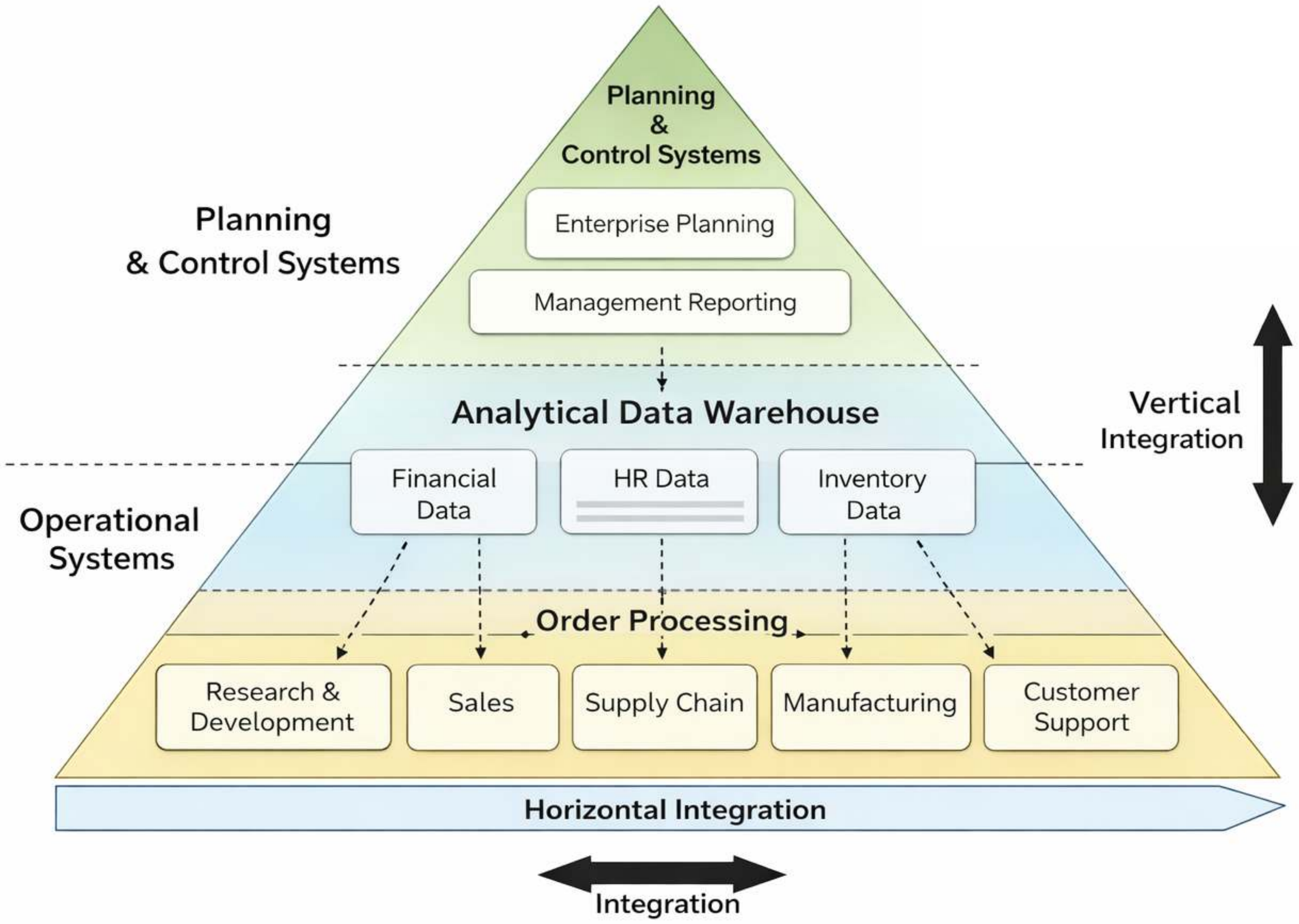
(2026-03-30)

🎯 **Learning objectives**

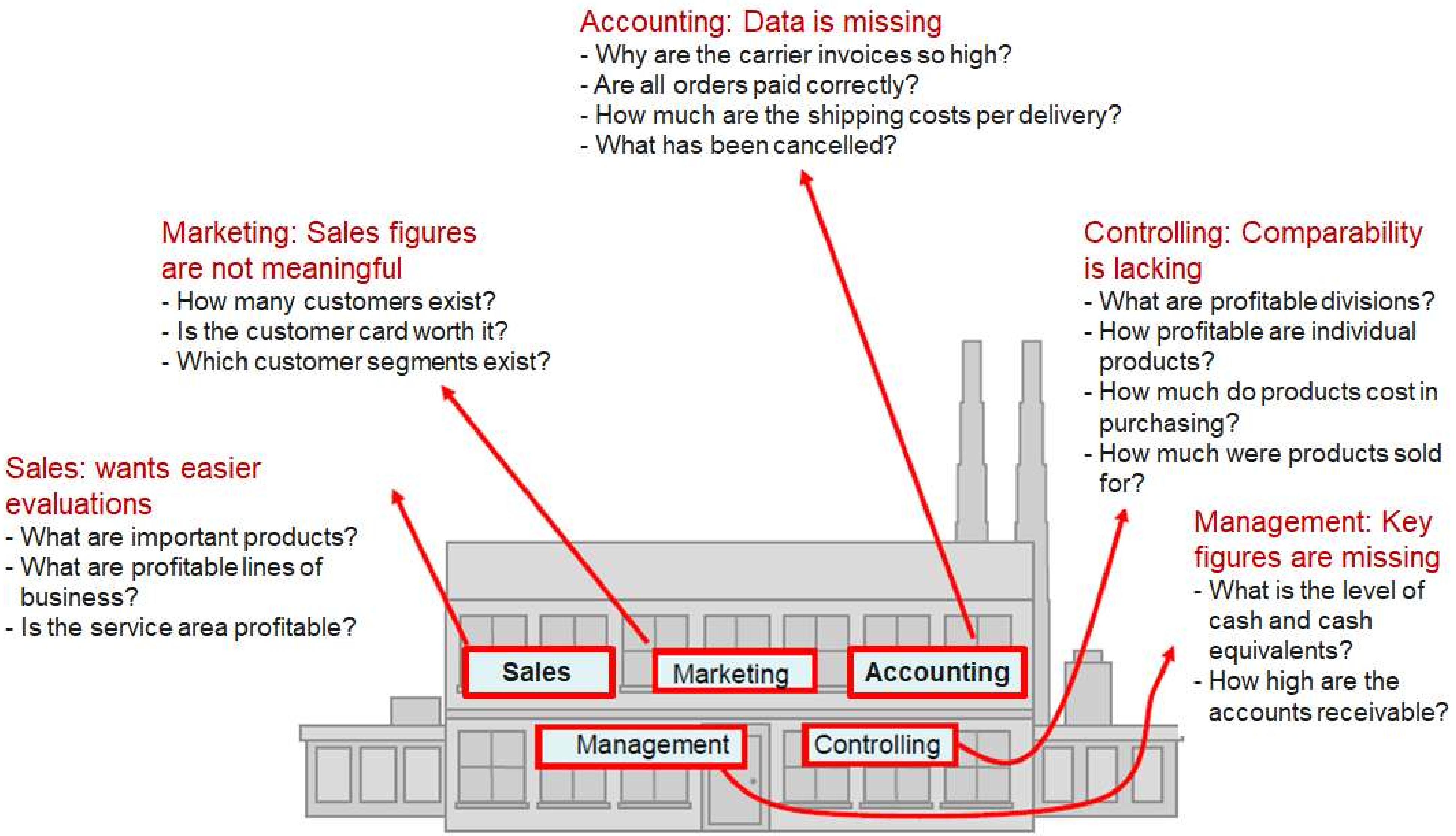
- **Outline** the architecture of data warehouses.
- **Explain** dimensional modeling concepts and common schemata.
- **Design** fact and dimension schemata from business questions and operational data.

Data warehouse

Application systems pyramid



Typical descriptive questions

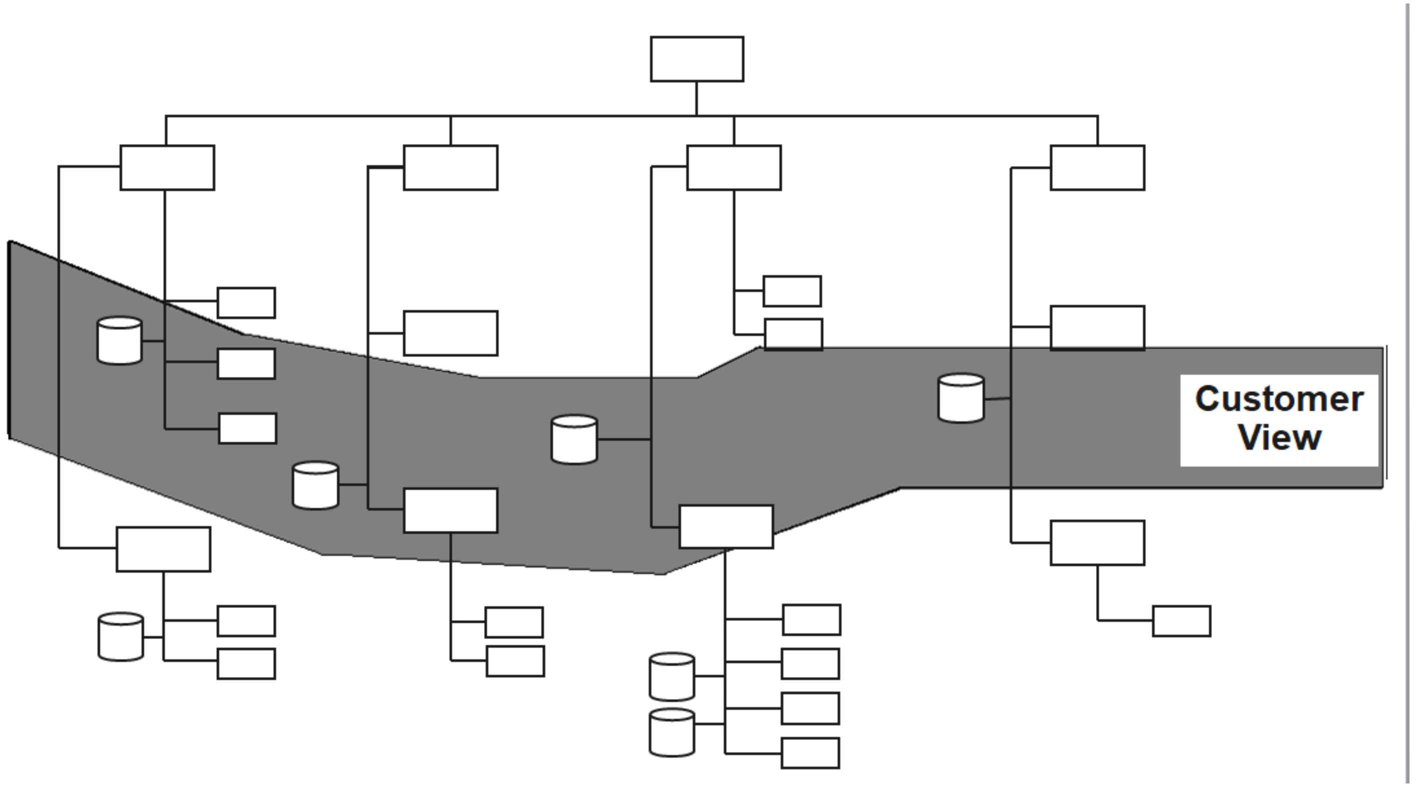


— Source: Alfred Schlaucher, Oracle, p. 15

Characteristics of operational and analytical databases

	Operational database	**Analytical database**
Target group	Operational employees	Management, Analysts
Access frequency	High	Slow to medium
Data volume per access	Low to medium volume	High volume
Required response time	Very short	Short to medium
Level of data	Detailed	Aggregated, processed
Queries	Predictable, periodic	Unpredictable, ad hoc
Focal period of time	Current	Past to future
Time horizon	1–3 months	Years or decades

Contradiction between operational data architecture and analytical information needs

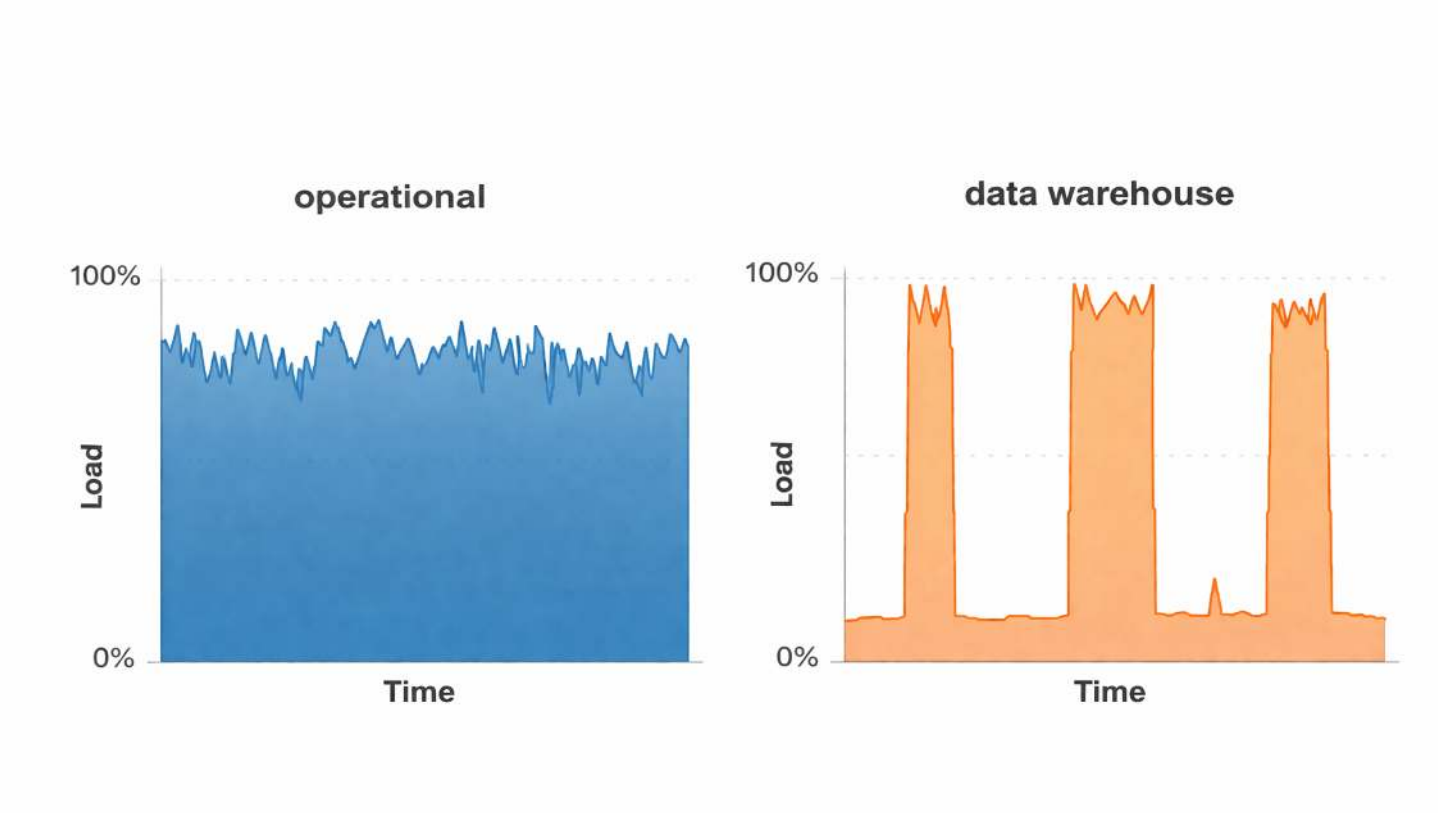


> Organizations structure operational data by departments and processes, but analytical questions require cross-departmental views.

— Source: Jung/Winter (2000): Data Warehousing Strategie: Erfahrungen, Methoden, Visionen, Berlin, p. 7.

Different intensities of use

The hardware load differs in operational application systems and data warehouse systems over time



=> A data warehouse relieves the operational systems

— Source: Inmon (2005), p. 23.

Definition: Data warehouse

The term Data Warehouse was coined by Inmon (1995), who defined it as follows:

> **“A data warehouse is a subject-oriented, integrated, time-variant and non-volatile collection of data in support of management’s decision-making process.”**

He defined the terms in the sentence as follows:

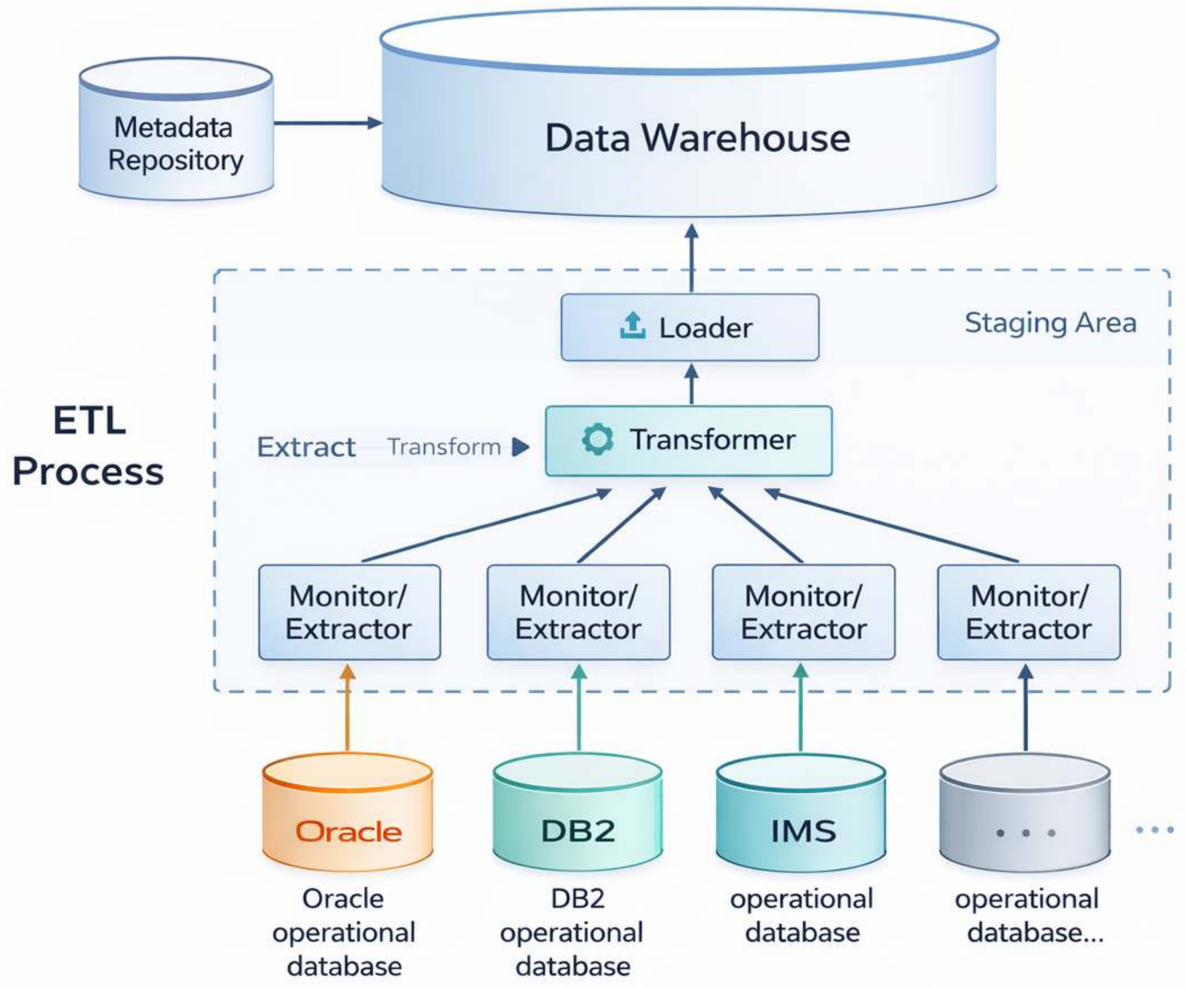
Subject-oriented: Data that gives information about a particular subject instead of about a company’s ongoing operations. Rather than reflecting fragmented departmental systems, it integrates data into consistent, cross-departmental subject areas.

Integrated: Data that is gathered into the data warehouse from a variety of sources and merged into a coherent whole.

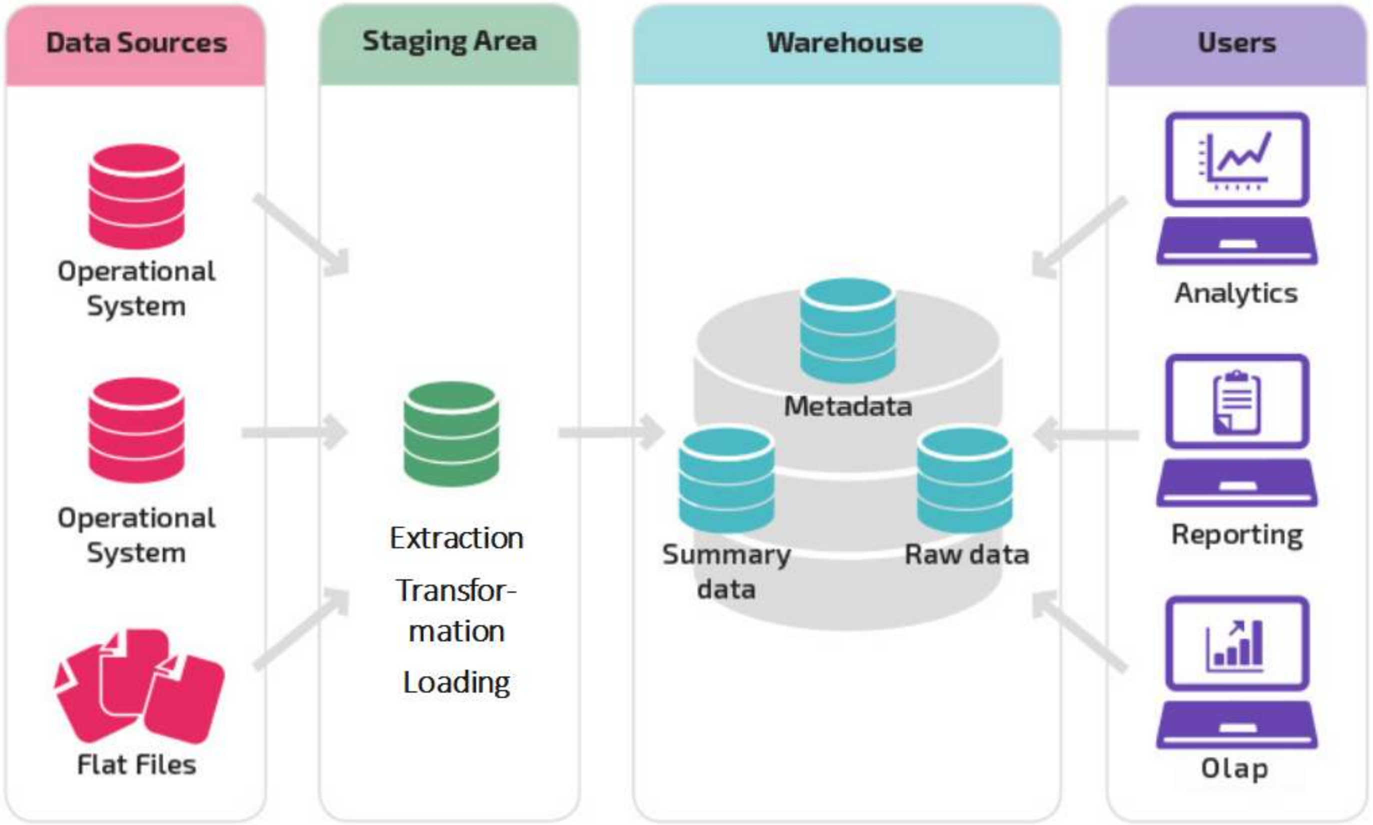
Time-variant: All data in the data warehouse is identified with a particular time period.

Non-volatile: Data is stable in a data warehouse. More data is added but data is never removed. This enables management to gain a consistent picture of the business.

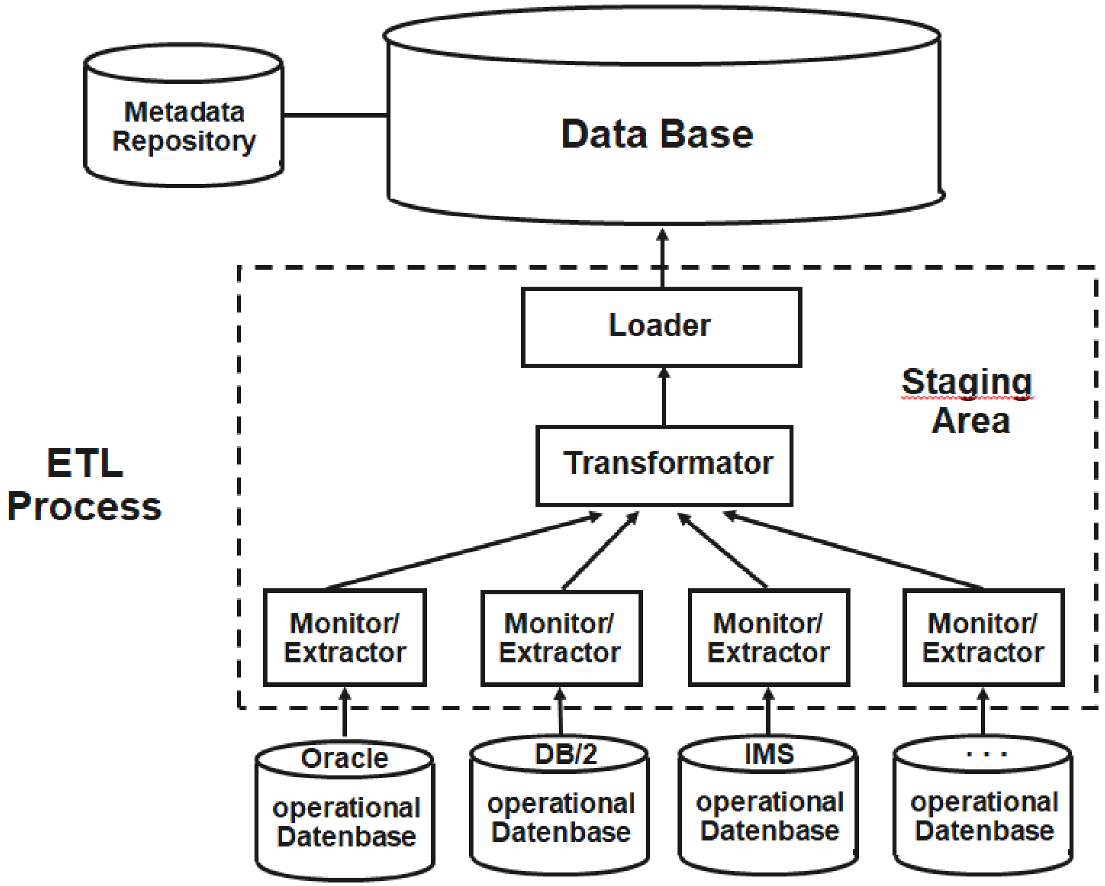
The data warehouse concept



Layers of a data warehouse system



Components of a data warehouse



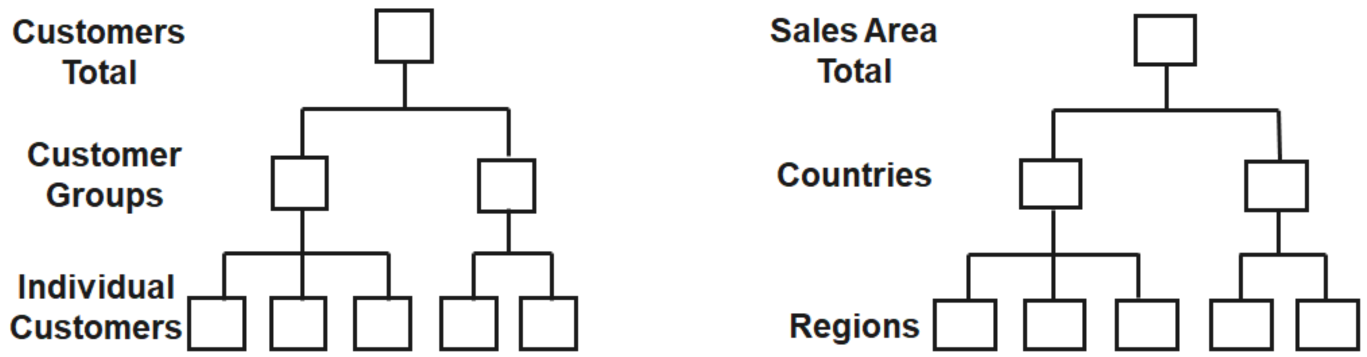
Structure of the database

The database represents the core of the data warehouse. It contains current and historical data from all relevant areas of the company. In contrast to the operational databases, the data here is not structured according to the business processes and functions of the company, but is aligned with the facts that determine the company. They are organized in such a way that they can be viewed from different dimensions.

Frequently used dimensions are:

- Product structure (e.g., product groups and individual products)
- Regional structure (e.g., region, country, continent)
- Customer structure (e.g., customer groups or segments, and individual customers)
- Time structure (e.g., month, quarter, year)
- Business parameters (e.g., sales, contribution margin, profit)
- Characteristics (e.g., target, actual, deviations)

Examples of aggregation levels in a data warehouse



ETL pipeline

ETL tools

The data in the data warehouse can normally only be accessed in a **read-only manner**. The only exception to this are ETL tools, which are responsible for transferring data from internal and external information sources. They automate the process of data acquisition from these sources.

The timeliness with which the transfer to the database is performed can be defined flexibly, e.g., within defined time intervals, immediately after each change, or at the explicit request of the user.

ETL programs include:

- **Monitors:** Detect and report changes in individual sources that are relevant to the data warehouse.
- **Extractors:** Select and transport data from the data sources into the workspace.
- **Transformers:** Unify, clean, integrate, consolidate, aggregate, and augment extracted data in the workspace.
- **Loaders:** Load transformed data from the workspace into the data warehouse after the data retrieval process is complete.

Staging area

The staging area is a temporary cache that performs the ETL process. It includes the extraction, transformation and loading of the data into the data warehouse.

Neither the operational systems nor the data warehouse are affected by the process. The data can be transferred incrementally from the operational systems without having to have a permanent connection to them. It is stored temporarily and only transformed when it is sufficiently complete.

It is then loaded en bloc into the data warehouse, so that no inconsistencies can occur here either.

If the ETL process is successful, the data is deleted from the staging area.

The metadata repository

Metadata supports data management and serves as a descriptor for an object that holds some data or information. In data warehouses, it is collectively organized in a catalog called the metadata repository.

Metadata is classified into three types:

- **Technical metadata:** Provides information about the structure of data, where it resides, and other technical details related to locating data in its native database.
- **Business metadata:** Describes the actual data in business-understandable terms. It can provide insights into the type of data, its origin, definition, quality, and relationships with other entities in the data warehouse.
- **Process metadata:** Stores information about the occurrence and outcomes of operations taking place in the data warehouse.

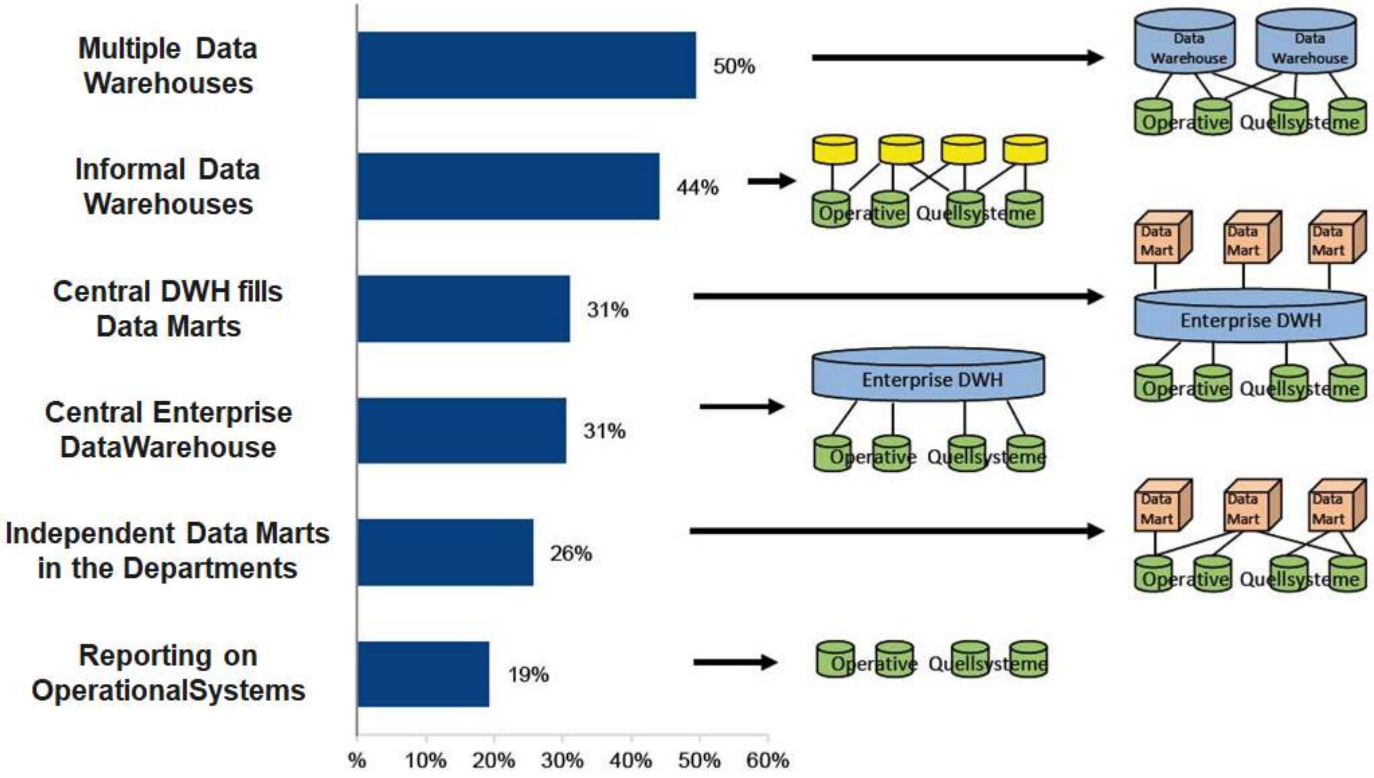
Data warehouse and data mart

A single-subject data warehouse is typically referred to as a Data Mart, while Data Warehouses are generally enterprise in scope.

Comparison: Data warehouse and data mart

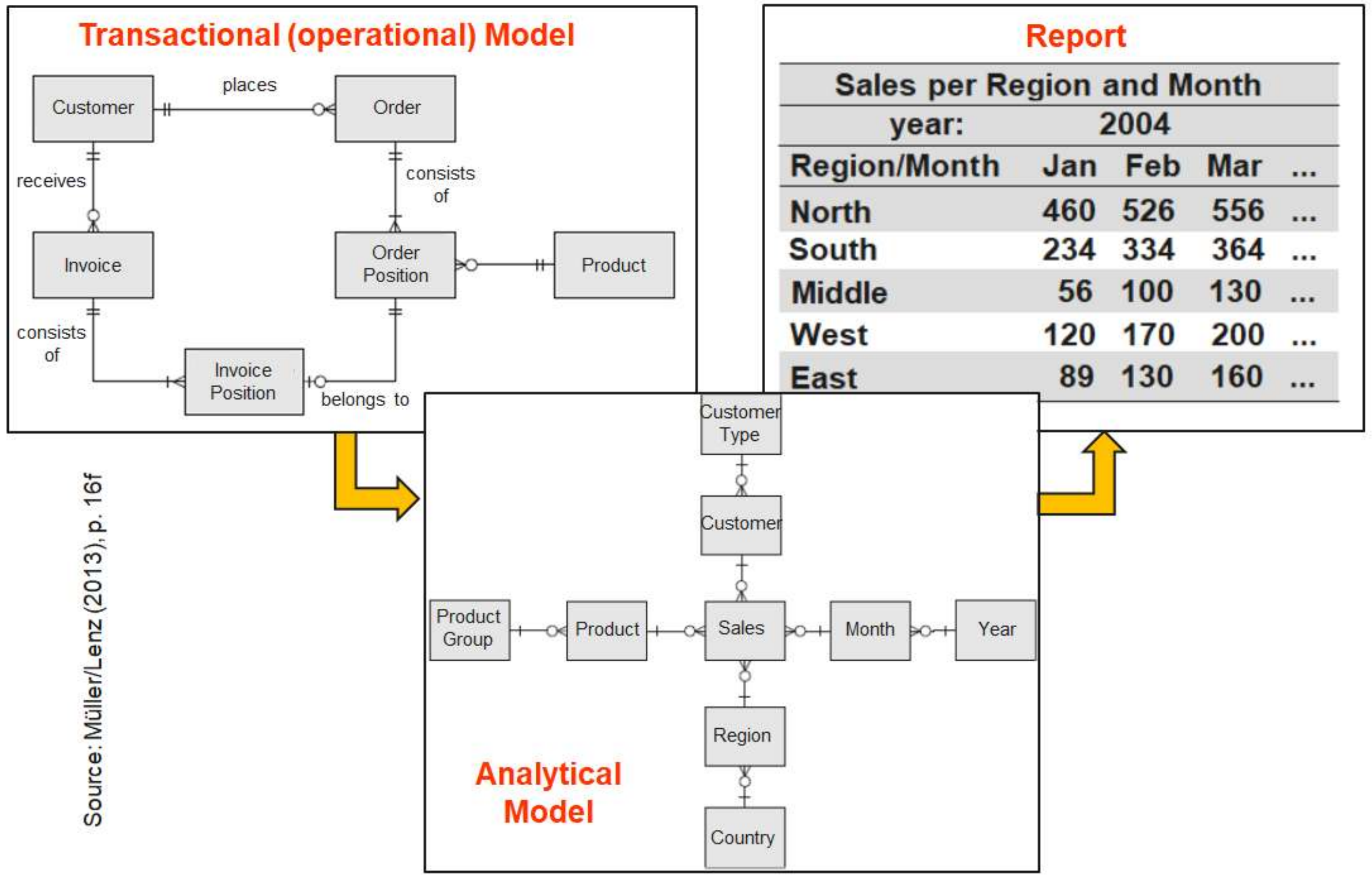
	Data mart	**Data warehouse**
Application	Division/Department	Company
Data design	Optimized	Generalist
Data volume	Gigabyte scale	Terabyte scale
Origin	Task oriented	Data model oriented
Requirements	Specific	Versatile

Architectural variants of data warehouses in practice

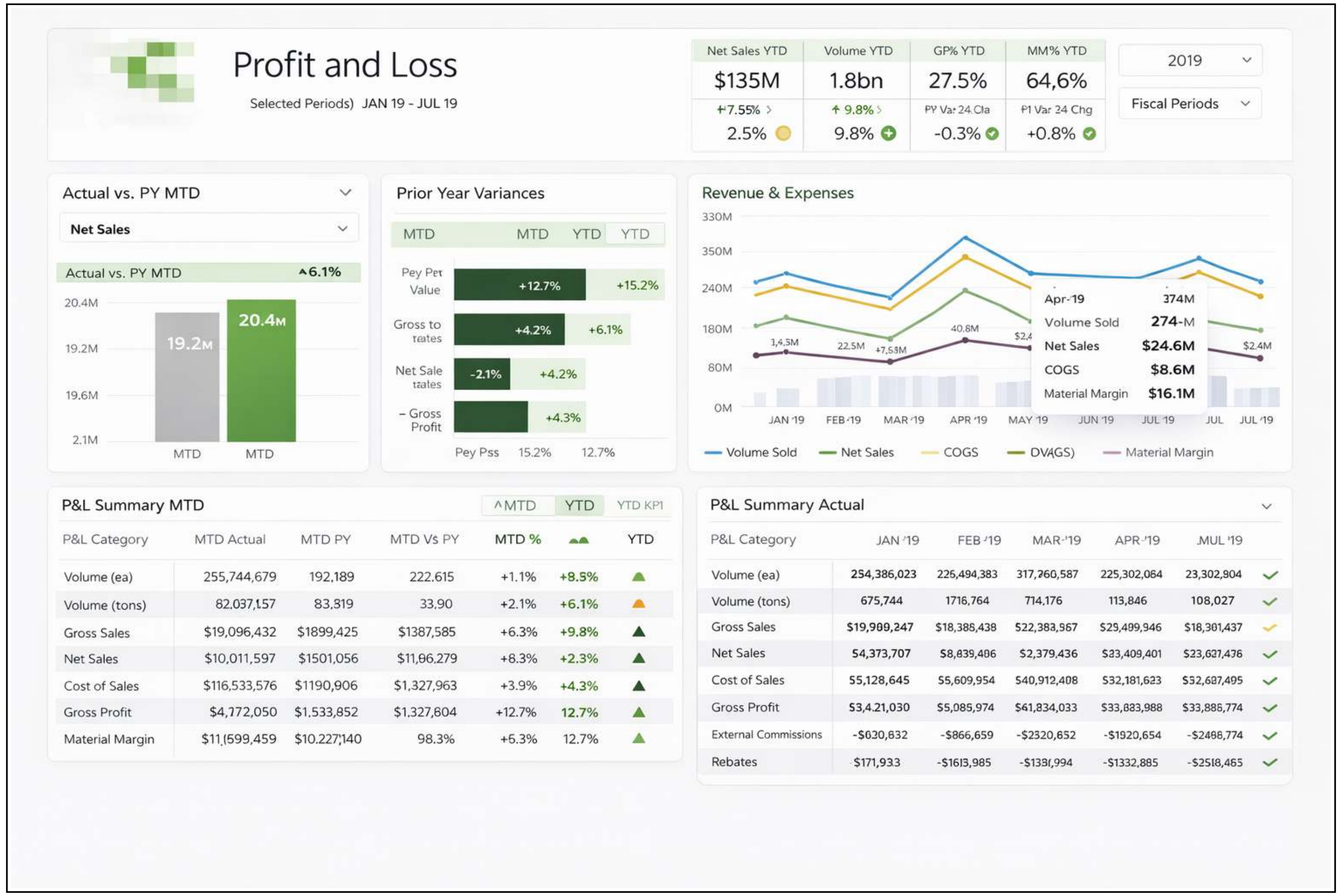


Dimensional modeling

From transactions to reports

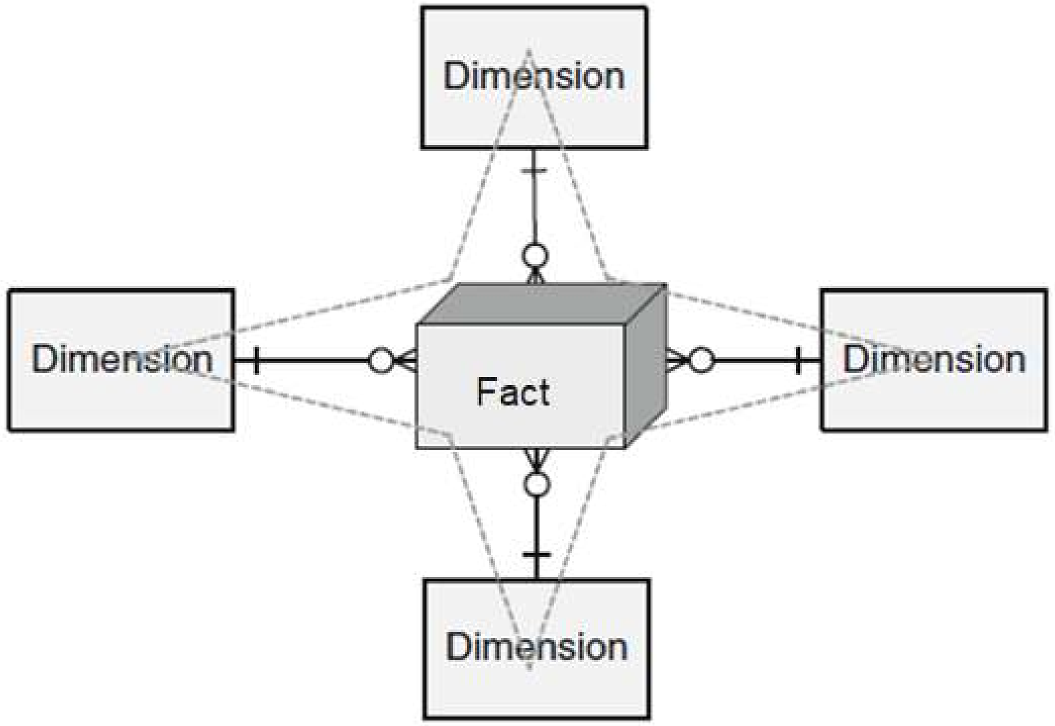


Example of a dashboard



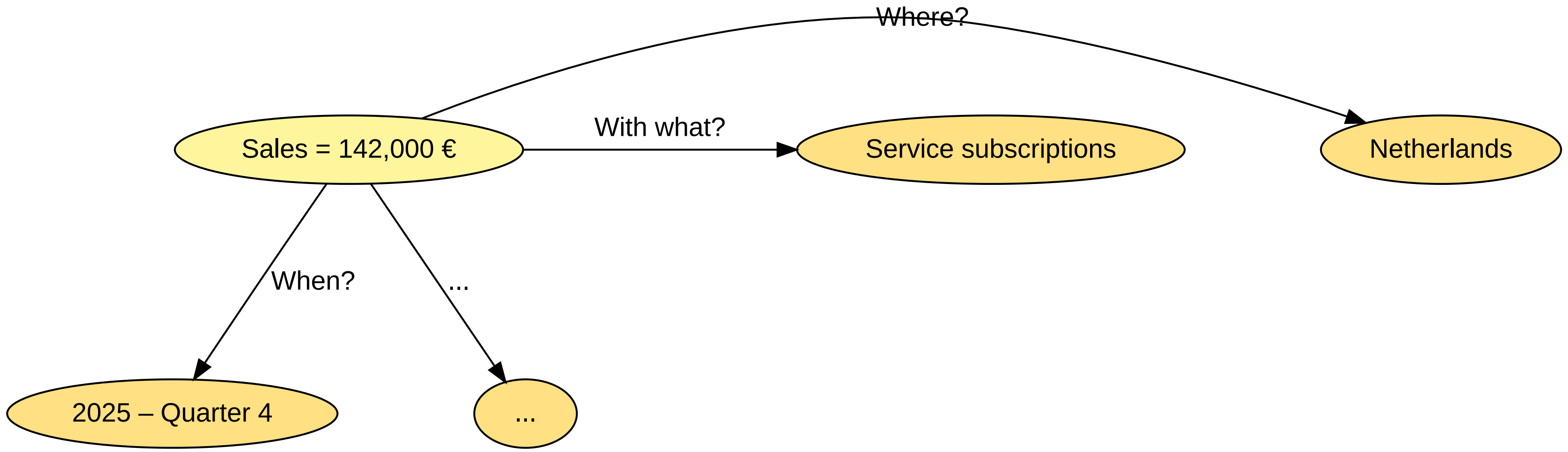
Star schema (I)

The Star schema has a central fact table and exactly one dimension table for each dimension.

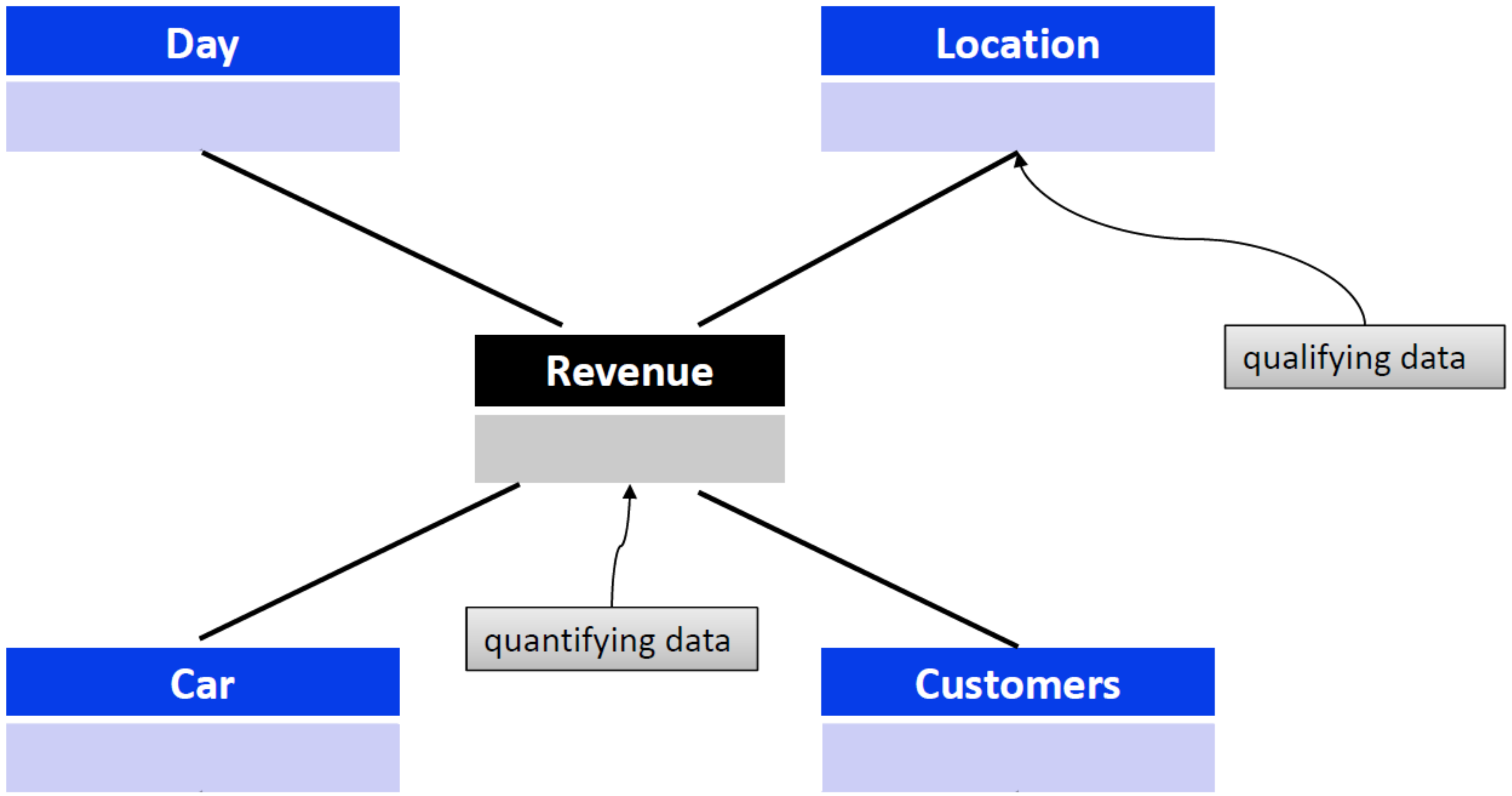


The cardinality between the fact table and a dimension table is n:1, i.e. a fact (e.g. sales: 67,000 euros) is described by one dimension expression for each dimension (e.g. time: 1st quarter, region: Hesse, product group: books). A dimension characteristic (e.g. product group: books) is associated with 0, 1 or n facts.

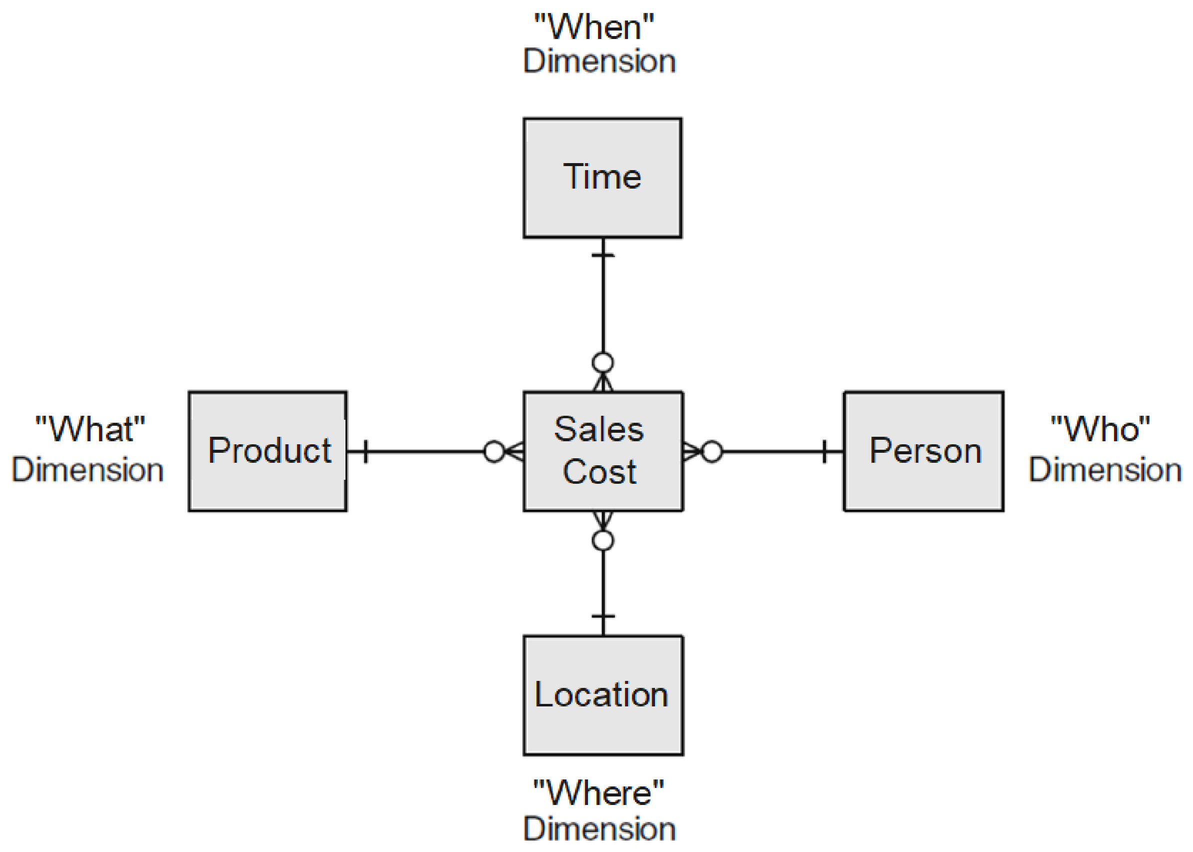
Star schema (II)



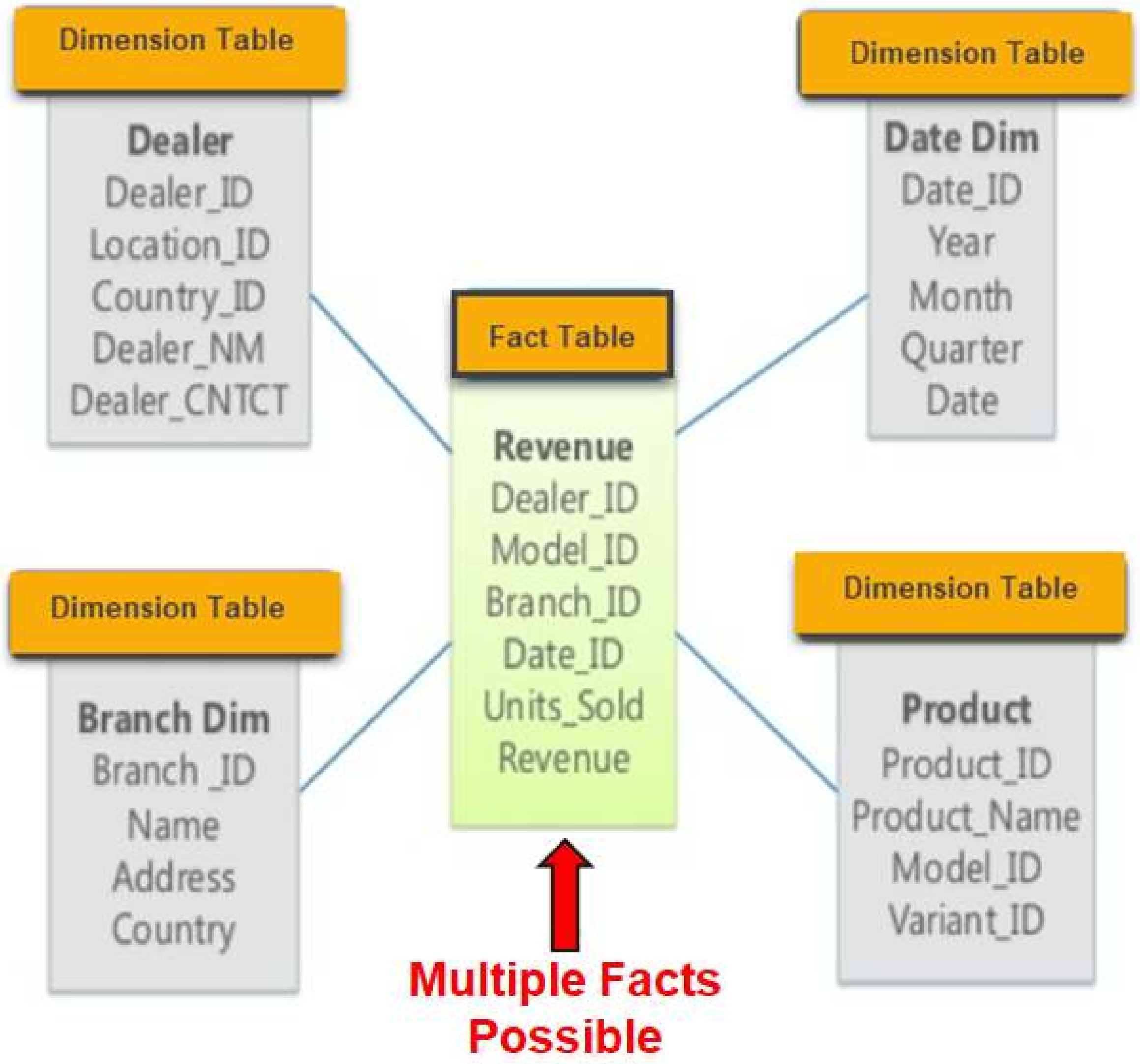
Star schema (III)



Star schema table design (I)



Star schema table design (II)



— Source: https://www.guru99.com/star-snowflake-data-warehousing.html

Star schema example

F (Sales)

Agency_ID	CustomerID	CarID	Revenue
HE_001	45678	BMW0017	330
HE_002	42973	BMW0008	260
HE_004	81115	VW1704	160
BW_001	81115	FORD2231	890

D_1 (Agency)

Agency_ID	Ag_City	State	Country
HE_001	Frankfurt	Hesse	BRD
HE_002	Fulda	Hesse	BRD
HE_004	Bensheim	Hesse	BRD
BW_001	Mannheim	Baden-W.	BRD

D_2 (Car)

CarID	C-Company	CarClass
BMW0017	BMW	Luxury
BMW0008	BMW	Family
VW1704	VW	Convertible
FORD2231	Ford	Eco-friendly

D_2 (Customer)

CustomerID	CustomerType	CustomerGroup
45678	Premium	Prefered
42973	Silver	Prefered
81115	Regular	Normal

Queries on the star schema

Example:

In which years were the most cars purchased by female customers in Hesse in the 1st quarter?

```
select z.Year as Year, sum(v.Amount) as Total_Amount
from Branch f, Product p, Time z, Customer k, Sales v
where z.Quarter = 1 and k.gender = 'f' and p.Product_type = 'Car' and f.state = 'Hesse' \
    and v.Date = z.Date and v.ProductID = p.ProductID and v.Branch = f.Branchname \
    and v.CustomerID = k.CustomerID
group by z.Year
order by Total_Amount Descending;
```

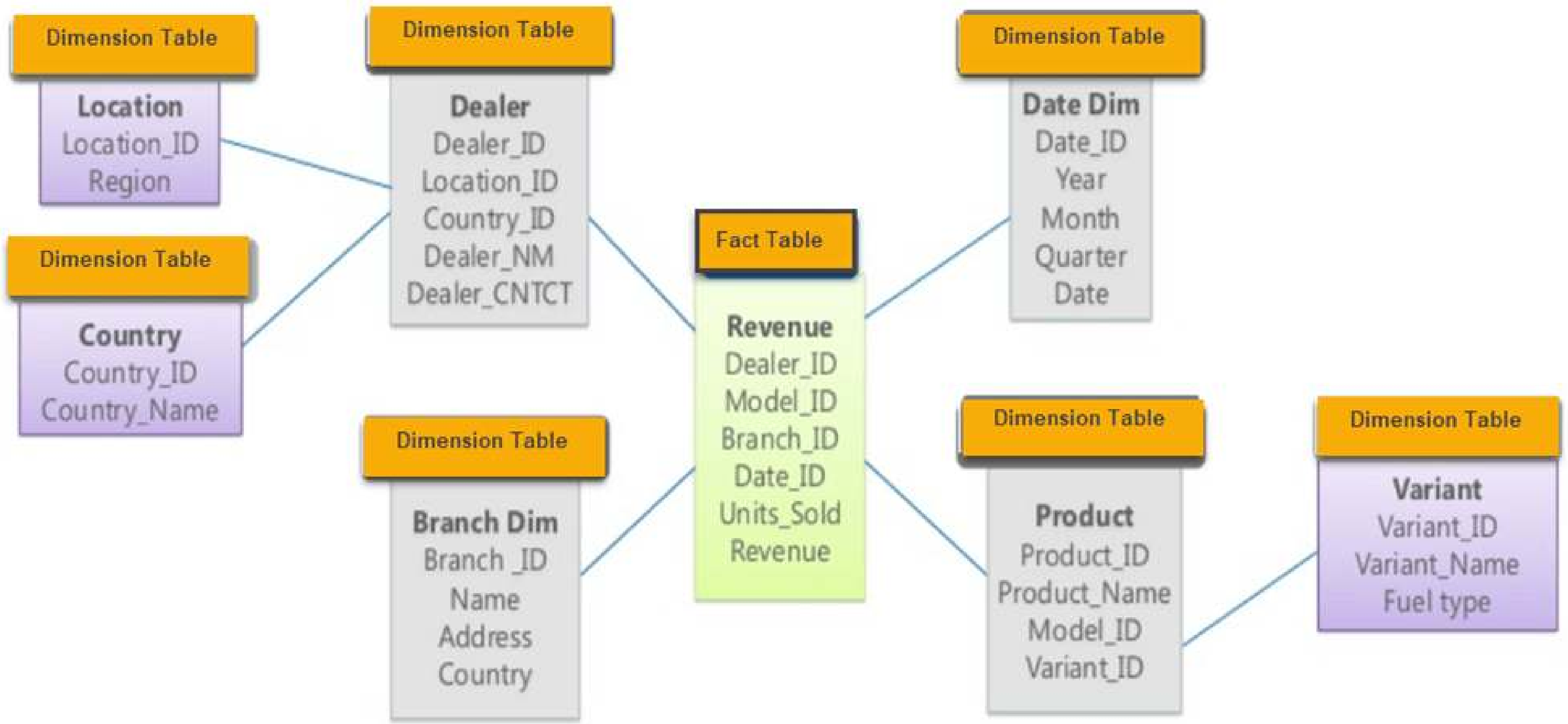
Year	Total amount
2004	745
2005	710
2003	650

Note: `FROM Branch f, Product p, Time z, Customer k, Sales v` uses **implicit join syntax**. All tables are combined first, and the **join conditions are defined in the `WHERE` clause** (e.g., `v.ProductID = p.ProductID`). This is equivalent to `JOIN ... ON`, but separates joins less clearly from filtering conditions.

Note: — Source: Hartung (2011), p. 37

Snowflake schema

In the Snowflake schema, the redundancies in the Star schema are resolved in the dimension tables and the hierarchy levels are each modeled with their own tables:



— Source: https://www.guru99.com/star-snowflake-data-warehousing.html

Star schema vs. snowflake schema

- Speed of query processing is in favor of Star scheme
- Data volume tends to be lower with Snowflake schema
- Snowflake schema is easier to change (maintenance)

Galaxy schema

The Galaxy schema contains more than one fact table. The fact tables share some but not all dimensions. The Galaxy schema thus represents more than one data cube (multi-cubes).

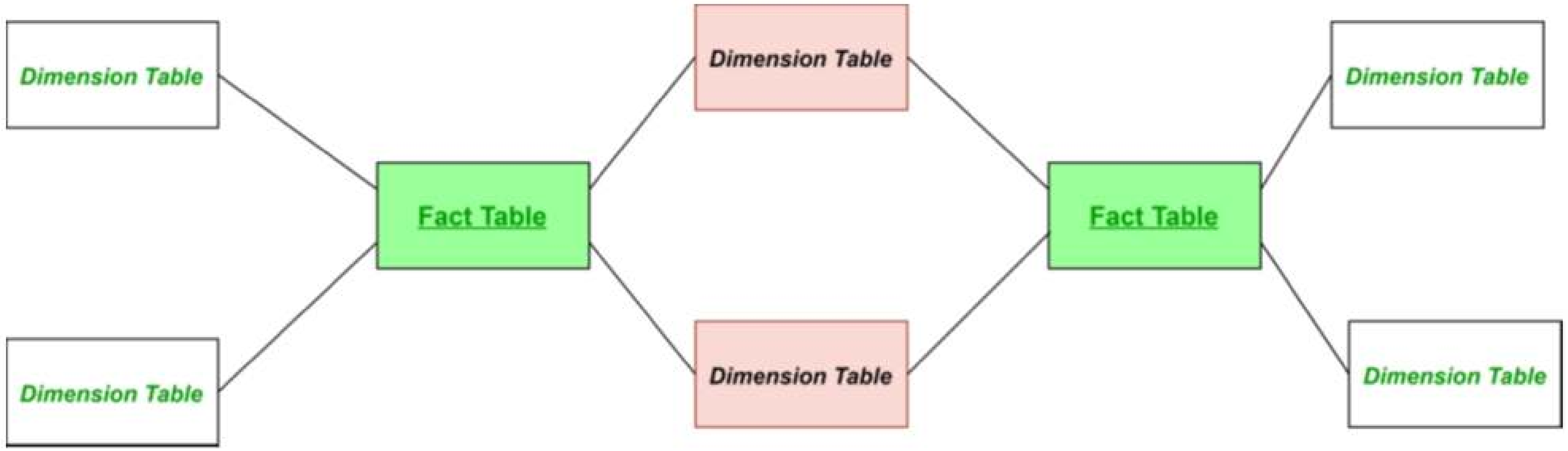
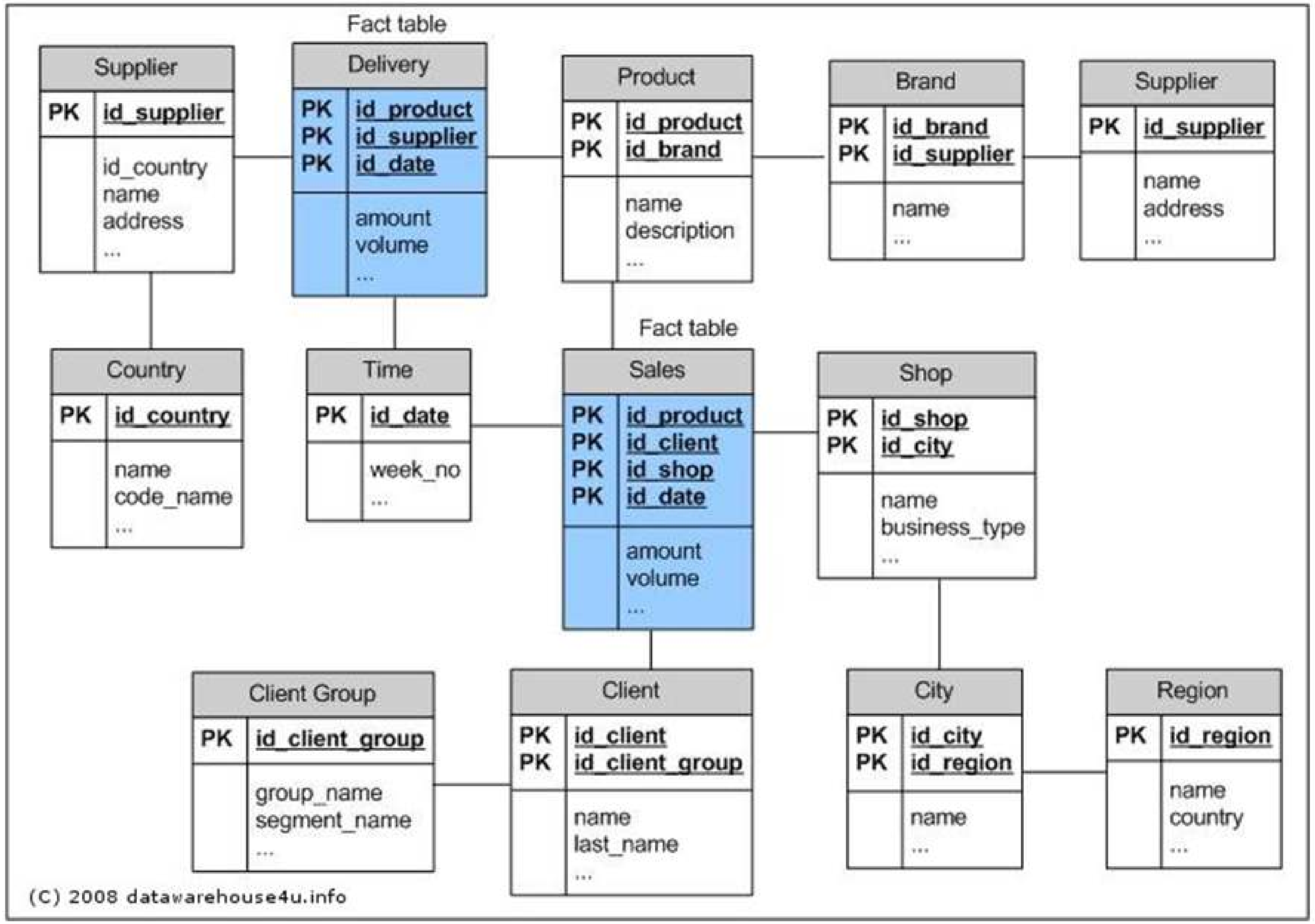


Table structure of the Galaxy schema



OLAP operations

Subject of OLAP

Online Analytical Processing (OLAP) is software technology focused on the analysis of dynamic, multidimensional data.

The goal is to provide executives with the information they need quickly, flexibly and interactively.

OLAP functionality can be characterized as dynamic, multidimensional analysis of consolidated enterprise data, which includes the following components:

- Calculations of key figures over different dimensions and aggregation levels
- Trend analysis over definable time intervals
- “What if” analyses
- Multidimensional visualization methods

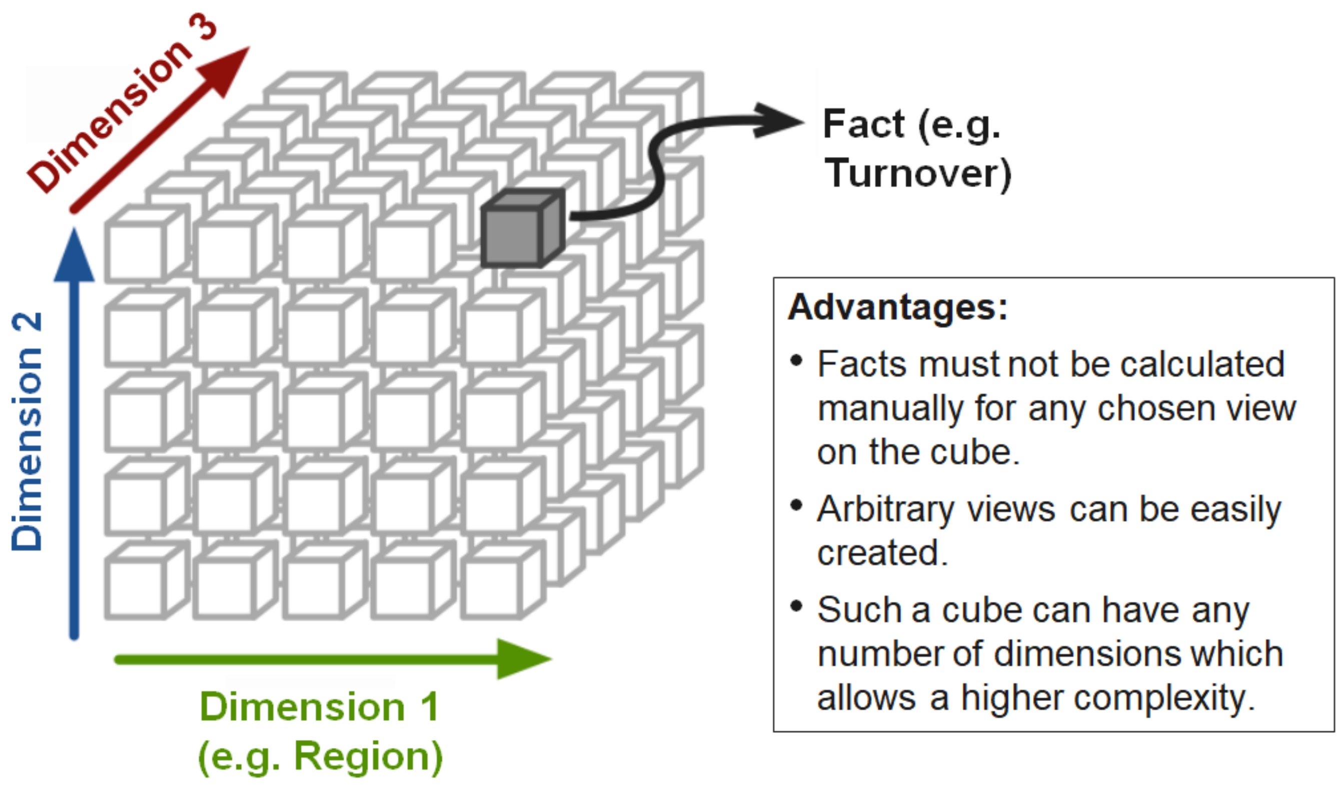
Facts and dimensions

Turnover		Product Group									
		Accessories		Desktops		Laptops		Monitors		Printers	
Region	Year	Actual	Target	Actual	Target	Actual	Target	Actual	Target	Actual	Target
Austria	2019	1435	1200	3444	3500	3728	4100	1808	1780	2001	2000
	2020	3109	3145	3453	3920	7558	9000	1481	1610	1285	1320
Brazil	2019	2812	3000	7294	7350	7529	8200	3636	3550	3895	3880
	2020	6079	5991	6972	6720	14780	15500	2952	3150	2591	2640
France	2019	2910	3150	7153	6650	7425	8200	3549	3560	4013	4000
	2020	6373	6840	6594	6720	15320	16300	2823	2870	2541	2580
Germany	2019	4349	4500	10392	10500	11355	11890	5663	5630	6061	6090
	2020	9370	9064	10130	10080	23103	24600	4290	4270	3877	3780
Great Britain	2019	3619	3600	9034	9100	8900	8610	4440	4440	4895	5020
	2020	8218	7964	8432	8680	19480	18900	3536	3500	3227	3120
Italy	2019	2927	2850	6839	6650	7534	7790	3553	3550	3967	4000
	2020	6215	5991	6711	6720	15325	16300	2852	2800	2653	2700
Switzerland	2019	2977	3150	7061	6650	7551	8200	3683	3710	4066	4130
	2020	6163	5991	6589	6720	15162	14700	2837	2660	2556	2520
USA	2019	14651	14700	35613	35700	37595	37310	18391	18380	20154	20050
	2020	30900	30433	32913	32760	74830	73900	14137	14070	12991	13020

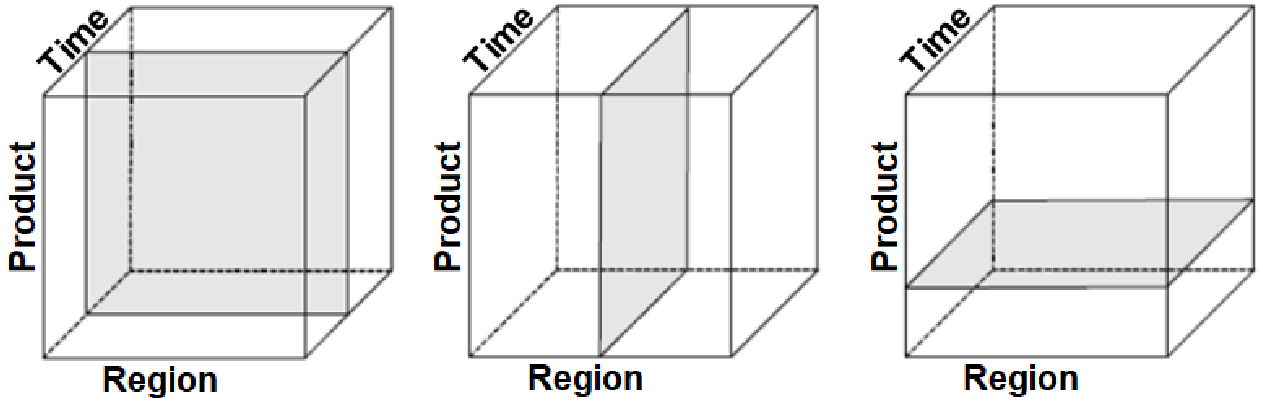
Qualifying Information (Dimensions)

Quantifying Information (Facts)

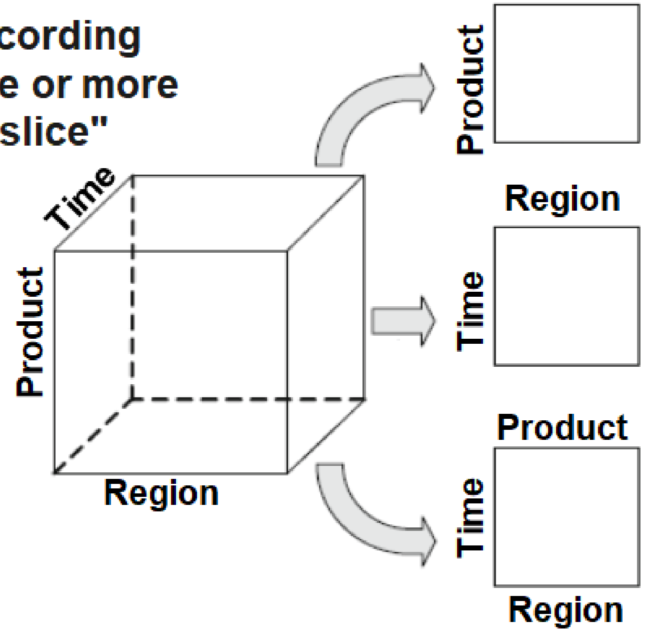
Organizing data as hypercube



OLAP operations: Slicing and dicing



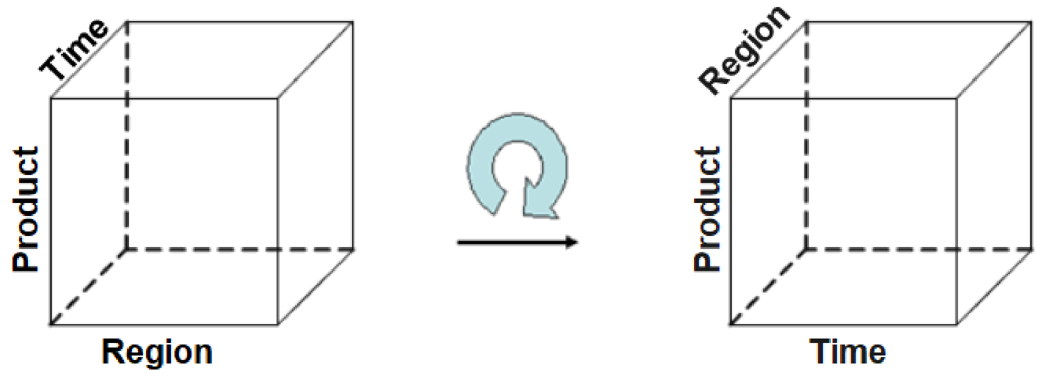
The **slicing** operator selects tuples from the cube according to the selection condition. The operator specifies one or more values for at least one dimension and thus slices a "slice" from the data cube.



The **dicing** operator selects a subset of the dimensions of the cube and uses it to cut out a smaller "sub-cube".

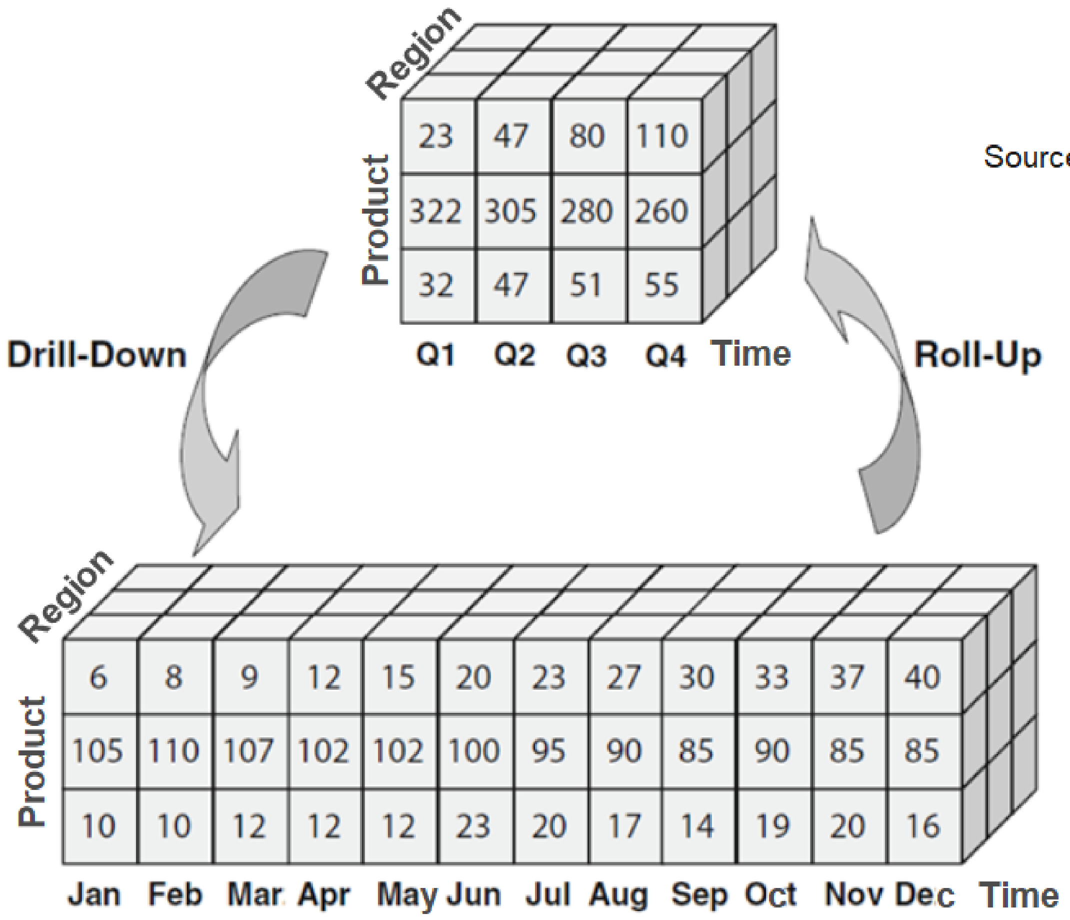
OLAP operations: Rotate

The rotate operator rotates the hypercube around one of its axes:



OLAP operations: Drill down and roll up

The roll-up operator corresponds to aggregation and goes from the specific up to the general in the dimensional hierarchy; conversely, drill-down goes as disaggregation from the general down to the specific in the hierarchy.



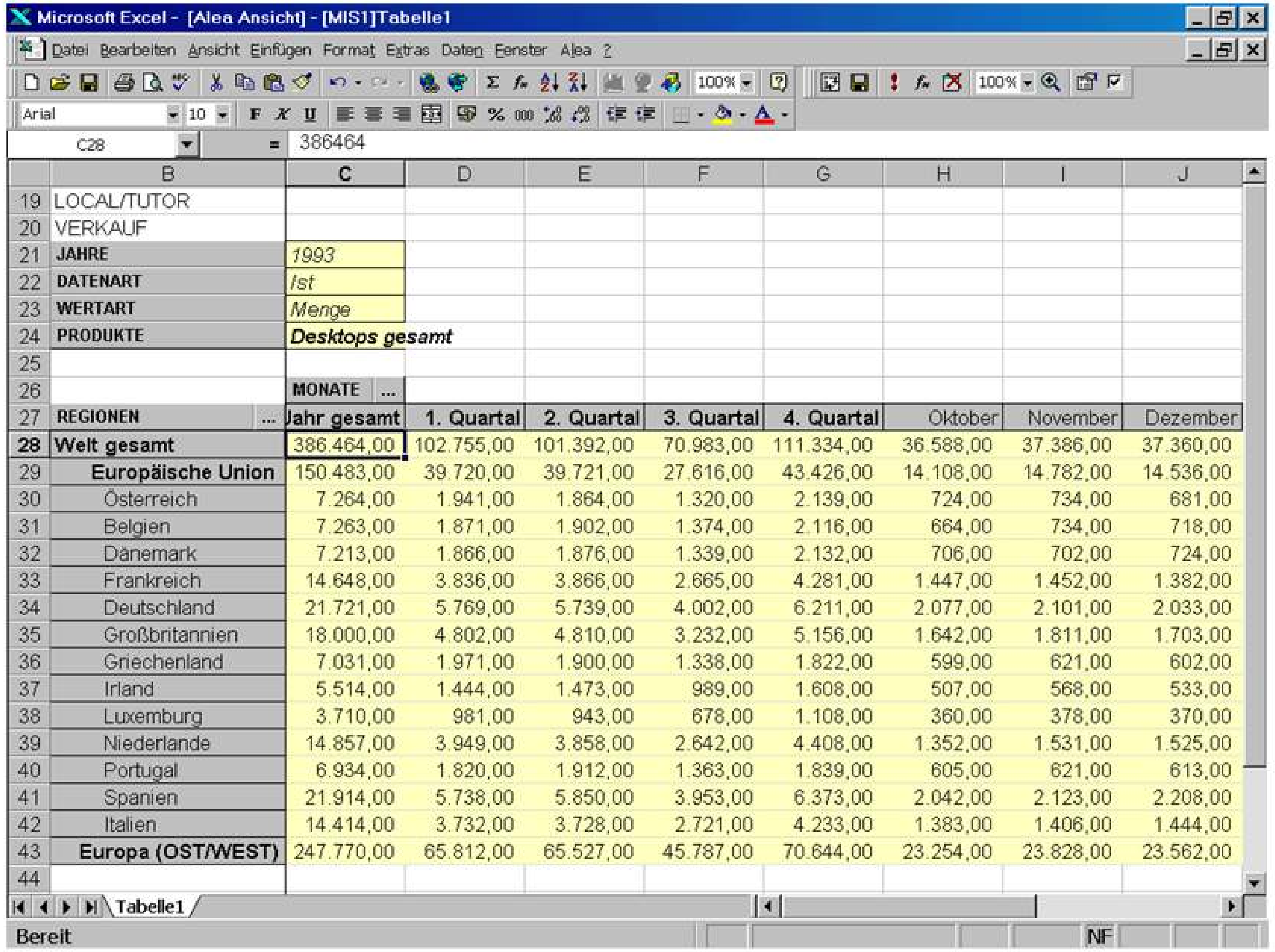
Source: Müller/Lenz (2013), p. 55

OLAP operations: Nest

The result of a nest operation physically represents a two-dimensional matrix. It is extended by displaying different hierarchy levels of one or more dimensions on one axis (column or row) in nested form. In the following example, three dimensions are displayed:

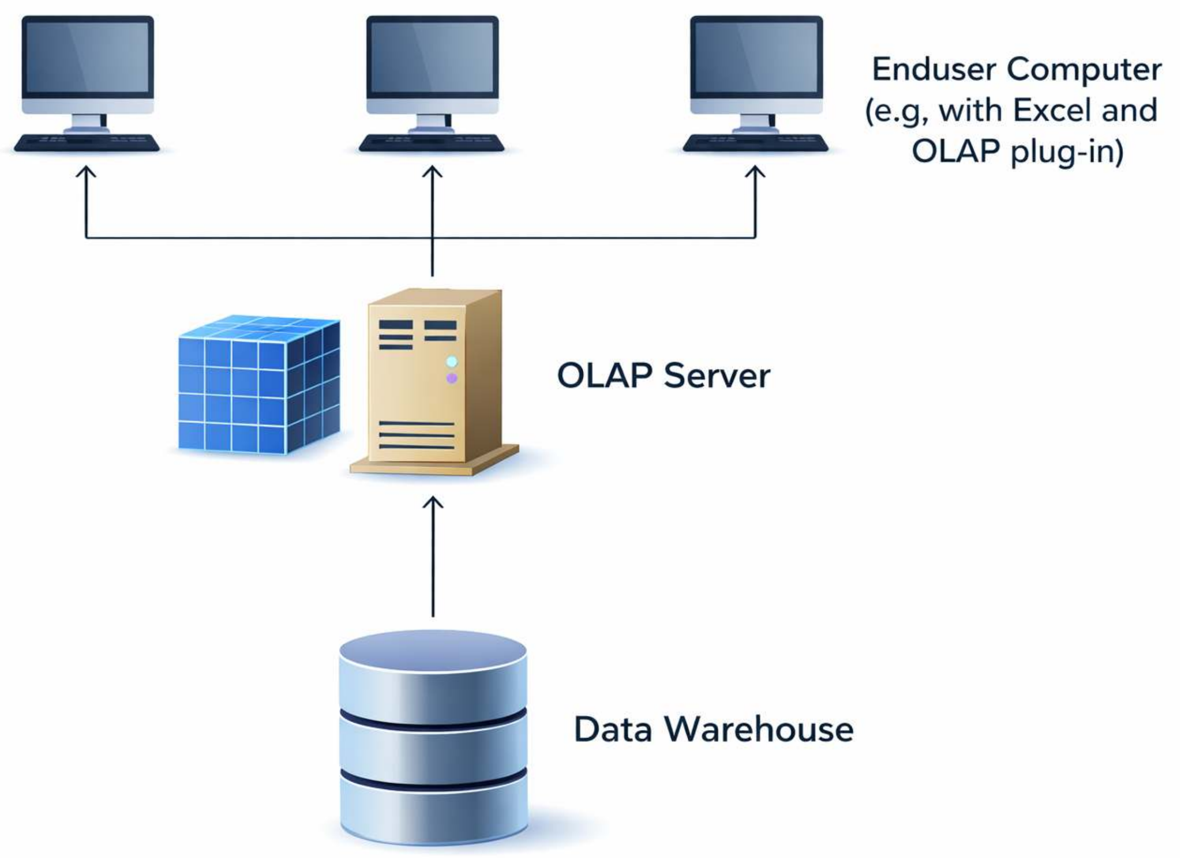
Turnover		Product Group					
Region	Year	Accessories	Desktops	Laptops	Monitors	Printers	Total
Austria	2019	1435	3444	3728	1808	2001	12416
	2020	3109	3453	7558	1481	1285	16886
Brazil	2019	2812	7294	7529	3636	3895	25166
	2020	6079	6972	14780	2952	2591	33374
France	2019	2910	7153	7425	3549	4013	25050
	2020	6373	6594	15320	2823	2541	33651
Germany	2019	4349	10392	11355	5663	6061	37820
	2020	9370	10130	23103	4290	3877	50770
Great Britain	2019	3619	9034	8900	4440	4895	30888
	2020	8218	8432	19480	3536	3227	42893
Italy	2019	2927	6839	7534	3553	3967	24820
	2020	6215	6711	15325	2852	2653	33756
Switzerland	2019	2977	7061	7551	3683	4066	25338
	2020	6163	6589	15162	2837	2556	33307
USA	2019	14651	35613	37595	18391	20154	126404
	2020	30900	32913	74830	14137	12991	165771

OLAP as Excel plug-in



	B	C	D	E	F	G	H	I	J
19	LOCAL/TUTOR								
20	VERKAUF								
21	**JAHRE**	*1993*							
22	**DATENART**	*Ist*							
23	**WERTART**	*Menge*							
24	**PRODUKTE**	***Desktops gesamt***							
25									
26		**MONATE** ...							
27	**REGIONEN** ...	**Jahr gesamt**	**1. Quartal**	**2. Quartal**	**3. Quartal**	**4. Quartal**	Oktober	November	Dezember
28	**Welt gesamt**	386.464,00	102.755,00	101.392,00	70.983,00	111.334,00	36.588,00	37.386,00	37.360,00
29	**Europäische Union**	150.483,00	39.720,00	39.721,00	27.616,00	43.426,00	14.108,00	14.782,00	14.536,00
30	Österreich	7.264,00	1.941,00	1.864,00	1.320,00	2.139,00	724,00	734,00	681,00
31	Belgien	7.263,00	1.871,00	1.902,00	1.374,00	2.116,00	664,00	734,00	718,00
32	Dänemark	7.213,00	1.866,00	1.876,00	1.339,00	2.132,00	706,00	702,00	724,00
33	Frankreich	14.648,00	3.836,00	3.866,00	2.665,00	4.281,00	1.447,00	1.452,00	1.382,00
34	Deutschland	21.721,00	5.769,00	5.739,00	4.002,00	6.211,00	2.077,00	2.101,00	2.033,00
35	Großbritannien	18.000,00	4.802,00	4.810,00	3.232,00	5.156,00	1.642,00	1.811,00	1.703,00
36	Griechenland	7.031,00	1.971,00	1.900,00	1.338,00	1.822,00	599,00	621,00	602,00
37	Irland	5.514,00	1.444,00	1.473,00	989,00	1.608,00	507,00	568,00	533,00
38	Luxemburg	3.710,00	981,00	943,00	678,00	1.108,00	360,00	378,00	370,00
39	Niederlande	14.857,00	3.949,00	3.858,00	2.642,00	4.408,00	1.352,00	1.531,00	1.525,00
40	Portugal	6.934,00	1.820,00	1.912,00	1.363,00	1.839,00	605,00	621,00	613,00
41	Spanien	21.914,00	5.738,00	5.850,00	3.953,00	6.373,00	2.042,00	2.123,00	2.208,00
42	Italien	14.414,00	3.732,00	3.728,00	2.721,00	4.233,00	1.383,00	1.406,00	1.444,00
43	**Europa (OST/WEST)**	247.770,00	65.812,00	65.527,00	45.787,00	70.644,00	23.254,00	23.828,00	23.562,00
44									

Architecture of an OLAP system



Data warehousing in modern analytics

DWH as foundation

- Codifies data preparation (ETL), transformations, and integration
- Centralizes **data access** and **metadata knowledge**

Evolution vs. r eplacement

- Accumulated logic and data understanding **favor incremental over greenfield evolution**

Data preparation

- Errors should be fixed **at the source systems**
- Transformations handled in **ETL processes**
- Duplicate handling depends on context (within vs. across datasets)

Role in modern analytics

- Serves as a **curated, reliable data layer**
- ML and advanced analytics typically operate **on top of the DWH**

Summary

- A **data warehouse** supports analytical decision-making by integrating data from multiple sources.
  It is **subject-oriented, integrated, time-variant, and non-volatile**, and separates analytical workloads from operational systems.
- The **ETL process** (Extract, Transform, Load) prepares data for analysis: data is extracted from sources, cleaned and integrated in a staging area, and loaded into the warehouse. **Metadata** ensures transparency and usability.
- **Dimensional modeling** organizes data for analysis:
  - **Facts** (e.g., sales) and **dimensions** (e.g., time, product, region)
  - Common schemata: **star** (simple, fast), **snowflake** (normalized), **galaxy** (multiple fact tables)
- **OLAP systems** enable interactive, multidimensional analysis: operations such as **slice, dice, drill-down, and roll-up** support flexible exploration of aggregated data.
- Overall, analytical data architecture transforms **operational data into structured information that can be efficiently queried** to support reporting, dashboards, and decision-making.

Survey: Session 3

https://forms.gle/dkwMZN4zPT7N3qbx5

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

References

Inmon, W. H. (1995). What is a data warehouse? *Prism Tech Topic*, *1*(1), 1–5.

Inmon, W. H. (2005). *Building the data warehouse*. John wiley & sons.